GE MRI

GRADIENT POWER AMPLIFIER (GPA) BASICS

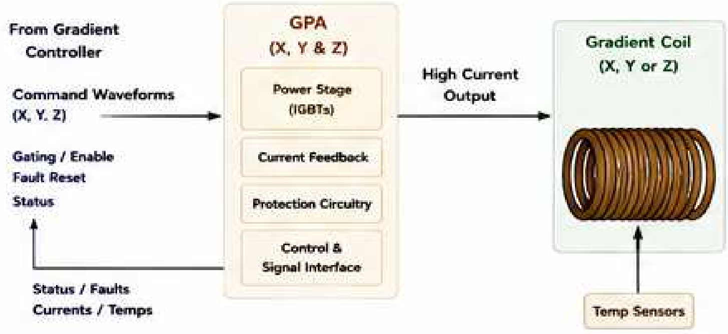
High Power. High Precision. Fast Switching. Cool Running.

The GPA drives the gradient coils with the commanded current waveforms that encode space.

WHY GPA PERFORMANCE MATTERS

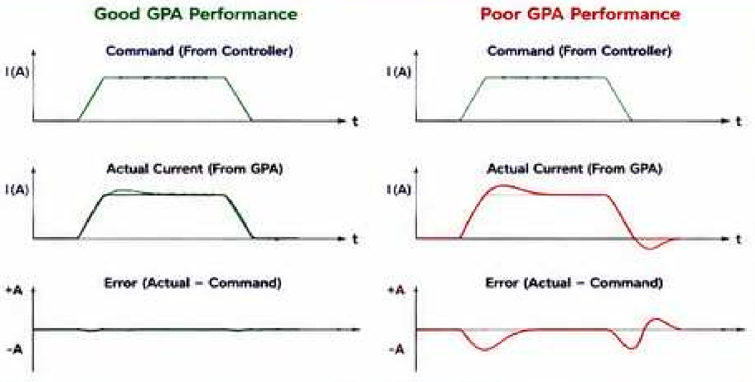
- Accurate current = accurate gradients = accurate images
- Fast response = short rise time = less distortion/ghosting
- Stable and cool = reliable scans and less downtime
- Fault protection = protects patient, system and hardware

1. GPA SYSTEM OVERVIEW



The GPA converts low-power commands into high-current, high-bandwidth waveforms that flow through the gradient coils.

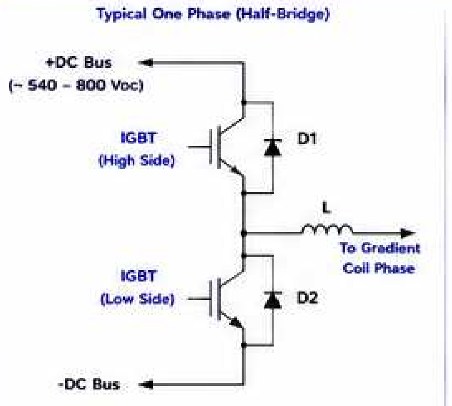
2. COMMAND vs ACTUAL CURRENT



Fast rise time, accurate amplitude, low error.

Slow rise, amplitude error, overshoot/undershoot.

3. GPA POWER STAGE – HOW IT WORKS (Simplified)

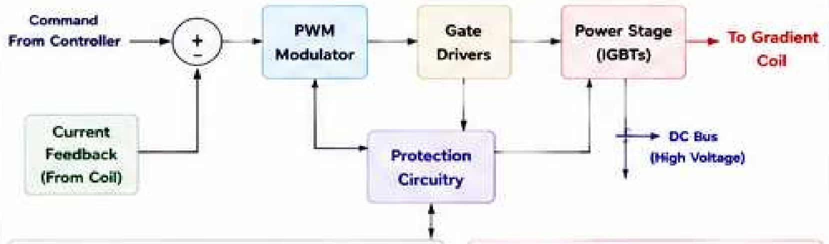


- IGBTs switch high current on/off rapidly.
- PWM (Pulse Width Modulation) controls average current to follow the command.
- Freewheel diodes (D1, D2) provide current path when switches turn off.
- LC filter (L) smooths the current.
- Current is bidirectional (±) to produce bipolar gradients.

Key Specs (Typical)

- Output Current (per axis): 300 – 600 A peak (system dependent)
- Output Voltage: ± 300 to ± 400 V (effective)
- Rise Time (10%-90%): 100 – 300 µs
- Current Ripple: < 1 – 2%
- Switching Frequency: 8 – 20 kHz

4. GPA BLOCK DIAGRAM (ONE AXIS)



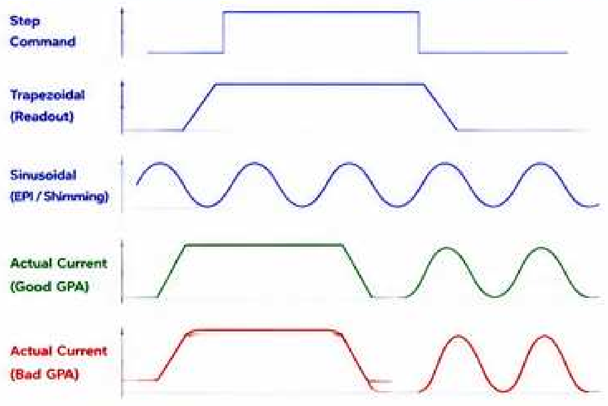
GPA functions in closed loop to force the actual current to follow the command.

- ✔ Compares command to actual current.
- ✔ Adjusts PWM to reduce error.
- ✔ Protects hardware and patients.

Protection Monitors

- Overcurrent (Instantaneous & RMS)
- Overvoltage (DC Bus)
- Overtemperature (Heatsink / IGBTs)
- Short to Ground / Phase
- Open Coil / Coil Mismatch
- DC Offset

5. WAVEFORMS YOU SHOULD KNOW



6. COMMON GPA FAULTS & WHAT THEY MEAN

Fault / Error	What It Means	Possible Causes	Check / Action
OVER CURRENT	Output current exceeds limit	Shorted coil, cable issue, slew rate too high, system fault	Check coil/cables, reduce slew, review sequence
OVER VOLTAGE	DC bus voltage too high	Regeneration, line issue, faulty bus regulation	Check line, bus caps, regenerative loads
OVER TEMP	GPA or heatsink too hot	Poor cooling, high duty cycle, fan failure	Check water flow, fans, clean filters, duty cycle
COIL OPEN / MISMATCH	Coil continuity problem or imbalance	Open coil, bad connector, axis mismatch	Measure coil resistance, check connectors
SHORT TO GROUND	Coil shorted to ground	Insulation breakdown, coil or cable fault	Insulation test, check cables and coil
DC OFFSET	Unwanted DC in current	Controller/GPA fault, feedback issue	Recalibrate, check offset adjust, service logs

7. INTERPRETING GPA STATUS (EXAMPLE)

GPA STATUS - AXIS X	
Output Current:	+12.3 A
RMS Current:	45.6 A
Command Amp:	50 %
DC Bus Voltage:	612 V
Heatsink Temp:	42 C
Water Temp In:	28 C
Water Temp Out:	32 C
Fault Status:	OK
Warnings:	NONE
Run Time:	12345 h

Monitor these values during scans and service:

- Output / RMS Current – actual load on axis
- Command Amplitude – how hard GPA is driving
- DC Bus Voltage – should be within spec
- Temperatures – ensure proper cooling
- Faults / Warnings – investigate immediately

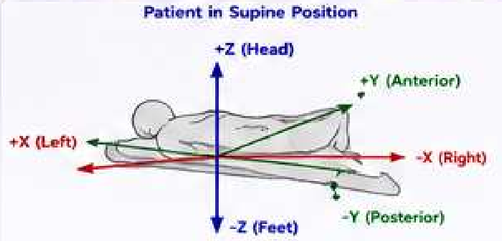
Trends over time help predict problems before they cause downtime.

8. TIPS FOR SERVICE ENGINEERS

- ✔ Always check GPA status first when you see image or gradient problems.
- ✔ Compare Command vs Actual Current (use Service / Monitoring tools).
- ✔ Look for amplitude error, delay, overshoot and ringing.
- ✔ Check water flow, temps and fans – overheating causes shutdowns.
- ✔ Verify coil resistance and continuity if faults point to hardware.
- ✔ Use test modes (Step, Ramp, Sin) to evaluate axis performance.
- ✔ Document baseline values after service.

A healthy GPA = clean waveforms, accurate currents, stable temps.

9. QUICK REFERENCE – AXIS IDENTIFICATION



Axis → GPA / Coil

- X → Readout (Left ↔ Right)
- Y → Phase (Anterior ↔ Posterior)
- Z → Slice (Head ↔ Foot)

Always verify axis orientation for the specific system.

ENGINEER NOTES

- Always document changes and results.
- Use test tools: GPA Monitor, Scope, Logs.
- Escalate if faults persist after checks.

ABBREVIATIONS

GPA	= Gradient Power Amplifier	RMS	= Root Mean Square
IGBT	= Insulated Gate Bipolar Transistor	TR	= Repetition Time
PWM	= Pulse Width Modulation	FOV	= Field Of View
DC	= Direct Current	EPI	= Echo Planar Imaging

SAFETY REMINDER

- High currents and voltages are present in GPAs.
- Do not remove covers or touch connections.
- Allow time for discharge. Follow lockout/tagout.
- Follow all GE safety procedures.

THE BOTTOM LINE

The GPA is the muscle of the MRI system.
It turns commands into precise gradient currents.
Keep it cool, keep it clean, keep it accurate.

GE HealthCare
Imagination at work